

Generation Interconnection System Impact Study Report

For Queue Project Q22-2140

Fairmont 138 kV

July 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2140. The Interconnection Customer proposes a 180.6 MW solar photovoltaic facility interconnecting at the Fairmont 138 kV substation in the Delaware area. The request was received on July 21, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 708 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,801,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Bond-Elk Grove Florin - Whitmore 138.0 kV
Contingency: Bond-Elk Grove Florin 345 kV/138 kV Transformer Bank
Loading: 116.6% (Rating: 465 MVA)
DFAX: 32.0%
Pre-project loading: 104.2%
2. Facility: Bond-Elk Grove Florin - Fairmont 138.0 kV Cir. 1
Contingency: Bond-Elk Grove Florin - Fairmont 138.0 kV Cir. 2
Loading: 106.2% (Rating: 806 MVA)
DFAX: 72.5%
Pre-project loading: 89.9%
3. Facility: Bond-Elk Grove Florin - Fairmont 138.0 kV Cir. 2
Contingency: Bond-Elk Grove Florin - Fairmont 138.0 kV line
Loading: 106.2% (Rating: 806 MVA)
DFAX: 72.5%
Pre-project loading: 89.9%
4. Facility: Sutherland Springs Substation - Hope Creek 138.0 kV
Contingency: Bond-Elk Grove Florin 345 kV/138 kV TX
Loading: 103.0% (Rating: 486 MVA)
DFAX: 32.0%
Pre-project loading: 91.1%
5. Facility: Hope Crk 345-138 kV Transformer Bank
Contingency: Bond-Elk Grove Florin Substation 345/138kV TX
Loading: 103.0% (Rating: 486 MVA)
DFAX: 32.0%
Pre-project loading: 91.1%
6. Facility: Stave Falls - Sutherland Springs 138.0 kV
Contingency: Bond-Elk Grove Florin 345 kV/138 kV Transformer Bank
Loading: 102.6% (Rating: 498 MVA)

DFAX: 32.0%
Pre-project loading: 91.0%

7. Facility: Stave Falls Substation - C W Tippy 138.0 kV
Contingency: Bond-Elk Grove Florin 345 kV/138 kV Transformer Bank
Loading: 102.2% (Rating: 509 MVA)
DFAX: 32.0%
Pre-project loading: 90.9%

8. Facility: Edgewater Station - C W Tippy 138.0 kV
Contingency: Bond-Elk Grove Florin 345/138kV Transformer Bank
Loading: 101.9% (Rating: 521 MVA)
DFAX: 32.0%
Pre-project loading: 90.8%

9. Facility: Edgewater Station - Whitmore 138.0 kV
Contingency: Bond-Elk Grove Florin 345/138kV Transformer Bank
Loading: 101.5% (Rating: 534 MVA)
DFAX: 32.0%
Pre-project loading: 90.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Whitmore - Bond-Elk Grove Florin 138.0 kV	Rebuild line with higher-capacity conductor	\$5,571,000	11% (\$637,000)
Bond-Elk Grove Florin - Fairmont 138.0 kV Cir. 1	Reconductor line	\$788,000	17% (\$137,000)
Bond-Elk Grove Florin - Fairmont 138.0 kV Cir. 2	Reconductor line	\$1,071,000	17% (\$187,000)
Sutherland Springs Substation - Hope Creek 138.0 kV Line	Reconductor line	\$3,630,000	11% (\$415,000)
Hope Crk 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$16,647,000	11% (\$1,904,000)
Stave Falls - Sutherland Springs 138.0 kV	Reconductor line	\$812,000	11% (\$93,000)
Stave Falls - C W Tippy Substation 138.0 kV	Reconductor line	\$767,000	11% (\$88,000)
C W Tippy - Edgewater Sta 138.0 kV Line	Reconductor line	\$1,401,000	11% (\$160,000)
Whitmore - Edgewater Sta 138.0 kV Line	Reconductor line	\$1,576,000	11% (\$180,000)
	Total allocated to project		\$3,801,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.